

The covariant-gradient Poincaré inequality on closed $C^{1,1}$ hypersurfaces

Candidate full answer to the open question in arXiv:2508.11109

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Status. Candidate full affirmative result, likely valid, requiring human review. For every compact connected embedded $C^{1,1}$ hypersurface of intrinsic dimension at least two, the covariant gradient is injective on tangential $W^{1,p}$ fields and satisfies the full Poincaré estimate, for every $1 < p < \infty$. This answers the exact regularity question following Lemma 3.2 of the source and lowers its weak L^2 Bochner–Laplace theory to $C^{1,1}$.

1 The source question and the answer

Let $\Gamma \subset \mathbb{R}^{d+1}$ be a compact connected embedded hypersurface without boundary, where $d \geq 2$. The source proves, under $\Gamma \in C^2$, that the covariant gradient ∇_Γ is injective on tangential fields and consequently that

$$\|v\|_{W^{1,p}(\Gamma)} \leq C \|\nabla_\Gamma v\|_{L^p(\Gamma)}, \quad v \in \mathbf{W}_t^{1,p}(\Gamma). \quad (1)$$

It then explicitly asks whether C^2 can be replaced by $C^{1,1}$.

In the proof above, we needed Γ to be of class C^2 . This requirement shall propagate to all well-posedness and regularity results in this paper, even in the L^2 -based setting. It would be interesting to determine if such requirement can be relaxed to Γ of class $C^{1,1}$. We leave this question open for future research.

We are in position to prove a Poincaré-type inequality in $\mathbf{W}_t^{1,p}(\Gamma)$.

Theorem 3.3 (Poincaré inequality). *Let $p \in (1, \infty)$ and assume Γ is of class C^2 . Then, there is a constant $C > 0$ depending only on Γ and p such that*

$$\|\nabla_\Gamma \mathbf{v}\|_{0,p;\Gamma} \geq C \|\mathbf{v}\|_{1,p;\Gamma}, \quad \forall \mathbf{v} \in \mathbf{W}_t^{1,p}(\Gamma). \quad (3.7)$$

Proof. This is a consequence of the Peetre–Tartar’s Lemma [26, Lemma A.38] applied to operators $A := \nabla_\Gamma \in \mathcal{L}(\mathbf{W}_t^{1,p}(\Gamma), \mathbb{L}^p(\Gamma))$ and $T := \text{id} : \mathbf{W}_t^{1,p}(\Gamma) \rightarrow \mathbb{L}^p(\Gamma)$ —which are respectively injective and compact by Lemma 3.2 and Remark 2.12—and inequality (3.3). $\square$

Figure 1: The open $C^{1,1}$ question and the ensuing Poincaré theorem in the current source PDF [1, p. 14].

The obstruction in the printed proof is precise. For a C^2 hypersurface, continuity of the shape operator gives a nonempty open set on which the Gauss–Kronecker curvature does not vanish. For $C^{1,1}$, the shape operator exists only almost everywhere, so that topological step cannot be copied. The replacement is the area formula for the Lipschitz Gauss map: it produces a *positive-measure* set on which the shape operator is invertible. A low-regularity curvature compatibility argument is then enough.

Theorem 1 (Affirmative answer). *Let $d \geq 2$, $1 < p < \infty$, and let $\Gamma \subset \mathbb{R}^{d+1}$ be a compact, connected, embedded $C^{1,1}$ hypersurface without boundary. If $v \in \mathbf{W}_t^{1,p}(\Gamma)$ and $\nabla_\Gamma v = 0$ almost everywhere, then $v = 0$ almost everywhere. Moreover, there is a constant $C = C(\Gamma, p)$ such that*

$$\|v\|_{W^{1,p}(\Gamma)} \leq C \|\nabla_\Gamma v\|_{L^p(\Gamma)} \quad \text{for every } v \in \mathbf{W}_t^{1,p}(\Gamma). \quad (2)$$

2 Three low-regularity lemmas

Write $\boldsymbol{\nu}$ for the outward unit normal, $\mathbf{P} = I - \boldsymbol{\nu} \otimes \boldsymbol{\nu}$, and $\mathbf{B} = \nabla_M \boldsymbol{\nu}$. On a $C^{1,1}$ hypersurface, $\boldsymbol{\nu}$ is Lipschitz and $\mathbf{B} \in L^\infty$; $\mathbf{B}$ is self-adjoint on the tangent space almost everywhere.

Lemma 2 (The Gauss map has nonsingular differential on a set of positive measure). *The set*

$$E := \{x \in \Gamma : \det(\mathbf{B}(x)|_{T_x \Gamma}) \neq 0\}$$

has positive d -dimensional surface measure.

Proof. The map $\boldsymbol{\nu} : \Gamma \rightarrow S^d$ is Lipschitz. It is onto. Indeed, fix $\omega \in S^d$ and maximize $x \mapsto x \cdot \omega$ on Γ . At a maximizer, ω is normal to Γ . Jordan–Brouwer separation makes Γ the boundary of its bounded complementary component, which lies in the supporting halfspace $x \cdot \omega \leq \max_\Gamma y \cdot \omega$; hence the outward normal there is ω .

Apply the area formula to the Lipschitz map $\boldsymbol{\nu}$ between the two d -manifolds. If $N(\boldsymbol{\nu}, \omega)$ is its multiplicity, then

$$\int_\Gamma J_\Gamma \boldsymbol{\nu} d\mathcal{H}^d = \int_{S^d} N(\boldsymbol{\nu}, \omega) d\mathcal{H}^d(\omega) \geq \mathcal{H}^d(S^d). \quad (3)$$

Rademacher’s theorem and the definition of the shape operator give $J_\Gamma \boldsymbol{\nu} = |\det \mathbf{B}|$ almost everywhere. The integral in (3) is positive, so E has positive measure. $\square$

Lemma 3 (A weakly parallel field is Lipschitz and has constant length). *If $v \in \mathbf{W}_t^{1,p}(\Gamma)$ and $\nabla_\Gamma v = 0$ almost everywhere, then $v \in W^{1,\infty}(\Gamma; \mathbb{R}^{d+1})$ and $|v|$ is constant almost everywhere on Γ .*

Proof. Work in a $C^{1,1}$ chart. Differentiating $v \cdot \boldsymbol{\nu} = 0$ weakly and using that the tangential part of every first derivative of v vanishes yields

$$\partial_i v = -(v \cdot \partial_i \boldsymbol{\nu}) \boldsymbol{\nu} = A_i v \quad \text{a.e.}, \quad (4)$$

where $A_i \in L^\infty$. If the current integrability exponent q is below d , Sobolev embedding first gives $v \in L^{q^*}$, where $q^* = dq/(d - q)$, and (4) then gives $v \in W^{1,q^*}$. Iteration reaches an exponent at least d in finitely many steps. At the endpoint $q = d$, use $W^{1,d} \hookrightarrow L^r$ for every finite r ; once $q > d$, Morrey gives $v \in L^\infty$. Equation (4) finally gives $v \in W^{1,\infty}$.

Now

$$\partial_i |v|^2 = 2v \cdot \partial_i v = -2(v \cdot \partial_i \boldsymbol{\nu})(v \cdot \boldsymbol{\nu}) = 0$$

almost everywhere. A Sobolev function with zero weak gradient is constant on the connected manifold Γ . $\square$

The next lemma records the only compatibility fact whose regularity needs care. It is the $C^{1,1}$ version of “a parallel field is annihilated by curvature.”

Lemma 4 (Weak Gauss compatibility). *Under the hypotheses of Lemma 3, for almost every $x \in \Gamma$ and all $X, Y \in T_x \Gamma$,*

$$(v \cdot \mathbf{B}Y) \mathbf{B}X - (v \cdot \mathbf{B}X) \mathbf{B}Y = 0. \quad (5)$$

Proof. Choose a local $C^{1,1}$ parametrization $F : U \rightarrow \Gamma$. The induced metric is $W^{1,\infty}$, its Christoffel matrices Γ_i belong to L^∞ , and the coordinate vector V of v belongs to $W^{1,\infty}$ by Lemma 3. Parallellity says

$$\partial_i V + \Gamma_i V = 0 \quad \text{a.e. on } U. \quad (6)$$

Commuting the two distributional first-order operators in (6) gives

$$(\partial_i \Gamma_j - \partial_j \Gamma_i + \Gamma_i \Gamma_j - \Gamma_j \Gamma_i) V = 0 \quad \text{in } \mathcal{D}'(U). \quad (7)$$

This calculation is legitimate at exactly this regularity: products of an L^∞ coefficient with V or ∂V are integrable, and a term such as $(\partial_i \Gamma_j)V$ is defined by the Leibniz identity $\partial_i(\Gamma_j V) - \Gamma_j \partial_i V$.

For completeness, the Gauss identity also survives at this regularity. If $g_{ij} = \partial_i F \cdot \partial_j F$ and $h_{ij} = \partial_{ij} F \cdot \boldsymbol{\nu}$, then the coordinate expansion of the distributional curvature gives

$$\langle R(\partial_i, \partial_j) \partial_k, \partial_l \rangle = h_{il} h_{jk} - h_{ik} h_{jl} \quad \text{a.e.} \quad (8)$$

The apparent third derivatives of F cancel because distributional mixed partials commute; every remaining term is a product of L^∞ second derivatives. Thus the curvature distribution in (7) is represented by the L^∞ tensor in (8). Consequently $R(X, Y)v = 0$ almost everywhere. Rewriting (8) with $h(X, Y) = \langle \mathbf{B}X, Y \rangle$ yields exactly (5). $\square$

3 Proof of the theorem

Proof of Theorem 1. Let $v \in \mathbf{W}_t^{1,p}(\Gamma)$ and $\nabla_\Gamma v = 0$. At almost every point $x \in E$ from Lemma 2, diagonalize the self-adjoint shape operator: $\mathbf{B}e_i = \kappa_i e_i$, where all $\kappa_i \neq 0$. Write $v = \sum_i v_i e_i$. Lemma 4, with $X = e_i$ and $Y = e_j$, gives for $i \neq j$

$$0 = \kappa_i \kappa_j (v_j e_i - v_i e_j). \quad (9)$$

Because $d \geq 2$, for every index one may choose a different index in (9); hence every $v_i = 0$. Thus $v = 0$ almost everywhere on the positive-measure set E . Lemma 3 says that $|v|$ is constant, so that constant is zero. This proves injectivity.

It remains to prove the estimate. The a.e. tangential derivative identity

$$\nabla_M v = \nabla_\Gamma v - \boldsymbol{\nu} v^T \mathbf{B} \quad (10)$$

and $\mathbf{B} \in L^\infty$ give

$$\|v\|_{W^{1,p}(\Gamma)} \leq C_0 (\|\nabla_\Gamma v\|_{L^p(\Gamma)} + \|v\|_{L^p(\Gamma)}). \quad (11)$$

If (2) failed, there would be tangential fields v_n with $\|v_n\|_{W^{1,p}} = 1$ and $\|\nabla_\Gamma v_n\|_p \rightarrow 0$. Rellich compactness on the compact Lipschitz manifold gives, after passing to a subsequence, $v_n \rightharpoonup v$ in $W^{1,p}$ and $v_n \rightarrow v$ in L^p . The limit is tangential and $\nabla_\Gamma v = 0$, so injectivity gives $v = 0$. Inserting v_n in (11) now makes its right side tend to zero, contradicting $\|v_n\|_{W^{1,p}} = 1$. $\square$

4 Consequences and exact scope

Corollary 5 (Weak L^2 Bochner–Laplace theory). *For a hypersurface as in Theorem 1, the bilinear form*

$$a(u, v) = \int_\Gamma \nabla_\Gamma u : \nabla_\Gamma v$$

is coercive on $\mathbf{H}_t^1(\Gamma)$. Hence for every $f \in (\mathbf{H}_t^1(\Gamma))'$ there is a unique $u \in \mathbf{H}_t^1(\Gamma)$ such that $a(u, v) = \langle f, v \rangle$ for all $v \in \mathbf{H}_t^1(\Gamma)$, and

$$\|u\|_{H^1(\Gamma)} \leq C \|f\|_{(\mathbf{H}_t^1(\Gamma))'}.$$

Proof. Take $p = 2$ in Theorem 1 and apply Lax–Milgram. $\square$

The same replacement also lowers the source’s weak $W^{1,p}$ Bochner–Laplace theorem to $C^{1,1}$. Here is the propagation audit. The closed-range consequence of (2) makes

$$\operatorname{div}_\Gamma : \mathbf{L}_t^p(\Gamma) \longrightarrow (\mathbf{W}_t^{1,p'}(\Gamma))'$$

surjective. Starting from Corollary 5, represent the datum as the divergence of an L^p tensor and use (10) to rewrite each ambient component as a scalar Laplace–Beltrami equation. Its only new coefficients are $\nu \in W^{1,\infty}$ and $\mathbf{B} \in L^\infty$. Scalar divergence-form $W^{1,p}$ estimates on the Lipschitz induced metric and the finite Sobolev bootstrap from Lemma 3 reach the desired exponent for $p > 2$; duality gives $p < 2$. This is precisely the source’s proof of its weak Bochner theorem, with “ $\mathbf{B}$ continuous” replaced by the sufficient statement “ $\mathbf{B} \in L^\infty$.”

The packet makes no claim that higher-order hypotheses can be lowered. In particular, the source’s $W^{m+2,p}$ regularity statements require separate derivatives of the geometric coefficients and retain their stated $C^{m+2,1}$ assumptions. The weak Stokes/Oseen/Navier developments use additional inf–sup and commutator steps; the geometric bottleneck resolved here disappears, but those long downstream arguments are not repackaged as new theorems here.

Remark 6 (Why $d \geq 2$ is essential). *For a closed $C^{1,1}$ curve, the unit tangent is a nonzero parallel tangent field for the intrinsic one-dimensional connection. Thus neither injectivity nor (2) holds in dimension one. In the proof above, this is exactly where the use of two distinct indices in (9) would fail.*

5 Verification and novelty boundary

The proof is qualitative and uses no numerical computation. Its four review-critical points are:

1. the support-plane proof that the outward Lipschitz Gauss map is onto;
2. the area formula identity $J_\Gamma \nu = |\det \mathbf{B}|$ almost everywhere;
3. the distributional commutator and Gauss identity for a $W^{2,\infty}$ parametrization;
4. the algebraic implication (9) when $d \geq 2$.

Each is stable under changes on null sets, so the loss of pointwise curvature continuity creates no hidden localization issue.

The current source is arXiv:2508.11109v3, dated 4 March 2026; the arXiv record says that v3 changes only metadata, and the open sentence remains in the source. The run indexes and bounded searches through 11 August 2026 for the exact arXiv id, the exact question, $C^{1,1}$ covariant-gradient Poincaré inequalities, weakly parallel fields, and Lipschitz Gauss-map arguments found no resolution of this question. The ingredients are classical, so the result could be implicit in low-regularity submanifold geometry; no priority claim is made.

Human-review recommendation. Independently verify Lemma 4 in one coordinate chart, especially the multiplication of the curvature distribution by the $W^{1,\infty}$ field, and verify the Gauss-map orientation at a support point. Subject to those checks, classify Theorem 1 as a full affirmative answer to the source’s explicit $C^{1,1}$ question.

References

- [1] G. A. Benavides, R. H. Nochetto, and M. Shakipov, *L^p -based Sobolev theory on closed manifolds of minimal regularity: Vector-valued problems*, arXiv:2508.11109v3 (2026).
- [2] H. Federer, *Geometric Measure Theory*, Springer, 1969; see the area formula for Lipschitz maps.
- [3] L. C. Evans and R. F. Gariepy, *Measure Theory and Fine Properties of Functions*, revised ed., CRC Press, 2015; see Rademacher’s theorem and the area formula.